

Exact Methods for the Job Sequencing and Tool Switching Problem

The branch-and-Benders-cut and branch-and-price campaigns

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Outline

- 1 The problem, in one slide
- 2 Campaign 1 — branch-and-Benders-cut (BBC)
- 3 Campaign 2 — configuration branch-and-price (BNP)
- 4 Datasets, protocol, and the honest reading
- 5 The joint diagnosis

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The problem and the one quantity that governs everything

Data. Jobs J ($n = |J|$), tools T ($m = |T|$), magazine capacity b ; job j needs $T_j \subseteq T$, $|T_j| \leq b$. $U = \bigcup_j T_j$ is the set of tools used.

Decide. A job order and a loading policy; each tool insertion costs 1. Minimise total switches Z^* .

- Fixed order $\Rightarrow$ loading is solved *exactly* by KTNS (keep tool needed soonest), $O(nm)$ [Tang–Denardo 1988]
- The joint problem is NP-hard [Crama et al. 1994]

The coverage bound $Z^* \geq |U| - b$

Every used tool beyond the first magazine load must be inserted at least once. This one number is the LP value of *every* natural configuration relaxation — and on **about half** the standard instances it *is* the optimum. Both campaigns live or die by it.

Two exact solvers, one diagnosis

BBC — branch-and-Benders-cut

Put the *order* in an integer master; settle the *loading* exactly in a KTNS subproblem. Cuts feed the order back the true switch cost.

BNP — configuration branch-and-price

Reason in *tool configurations* (magazine-fillings). Exponentially many, so generate them as columns of a fixed-size linear program.

- Both are *our* exact methods; both are benchmarked against the compact baselines LSS'04, Catanzaro-F4'15, and the flow model SSPMF'24.
- They fail for the *same* reason, and the campaign's job is to prove it: the relaxation collapses to $|U| - b$.
- Where $|U| - b$ is tight, both close at the root; where it is loose, both must search — and cannot.

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BBC: why Benders is the natural decomposition

The textbook precondition. A *hard* ordering problem wrapped around an *easy* inner problem (loading, solved by KTNS). Benders is built for exactly this.

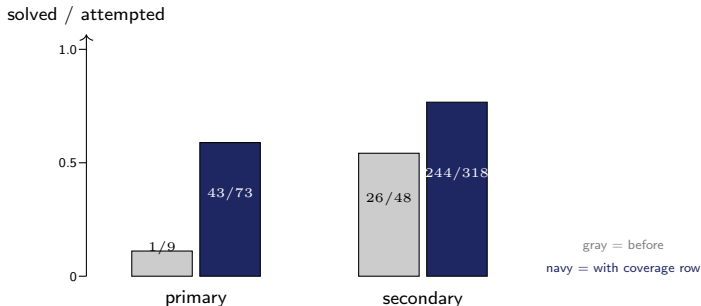
$\min \theta$

- s.t. x forms a Hamiltonian path through a depot (the order)
- $$\theta \geq \sum_{i,j} w_{ij} x_{ij}, \quad w_{ij} = \max(0, |T_i \cup T_j| - b) \quad (\text{pairwise bound})$$
- $$\theta \geq |U| - b \quad (\text{coverage row})$$
- $$\theta \geq \text{Benders optimality cuts from the KTNS dual} \quad (\text{exact})$$

- Fixed order $\Rightarrow$ loading is a *totally unimodular* LP whose value is KTNS $\Rightarrow$ its dual gives *exact* optimality cuts (never heuristic).
- Subtours separated exactly at integer candidates; 2^3 cut configurations ablated.

BBC: theory feeds the solver — the coverage row

Our Proposition $Z^* \geq |U| - b$ becomes *one master row* $\theta \geq |U| - b$. Its effect on the solve counts:



- On the 6-ring the proof closes at the root node, with no optimality cuts generated (443 were generated before the coverage row).
- The strongest configuration reaches 274/374 on the secondary suite. After the correction the eight cut configurations fall within a narrow band, so the earlier differences between them reflect the missing bound rather than the cut families.

BBC: the diagnosis — a pure proof problem

- 122 of 287 solves close at the root node; median solve time 0.05 s.
- In all 104 remaining time-limited runs the incumbent equals the known optimum — the optimum is found, but not certified within the time limit.
- No instance outside the bound-tight class is solved. Fractional Benders cuts are structurally inactive (no violated cut across the campaign), consistent with the subproblem value collapsing to the pairwise bound at fractional orders.

Shifted geometric-mean time on the 43 instances all four methods solve

	BBC (ours)	SSPMF'24	LSS'04	CATZ-F4'15
geom. mean time (primary)	1.0 s	0.1 s	333 s	616 s

On these instances root closure is about two orders of magnitude faster than the compact branch-and-bound models. The comparison is conditioned on the bound-tight instances BBC solves — which is the diagnosis itself.

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BNP: why the naive configuration model does not work

The SSP is a generalised TSP over configurations [Colares 2026]: visit one configuration serving each job, pay for tools that change.

- As first stated its *subtour elimination is invalid*; the exact correction needs **exponentially many** constraints (one per subset of configurations).
- Miller–Tucker–Zemlin (MTZ) repair: one potential *per configuration* is exact but exponential; aggregating to *one potential per job* is polynomial but **no longer exact**.

Why the aggregated MTZ fails (job domination)

If $T_i \subseteq T_j$, every configuration serving j also serves i ; a transition between two such configurations pins $\{i, j\}$ at both ends and forces their potentials into a 2-cycle no order can satisfy. A 4-job, $b = 3$ instance has optimum 1 but the aggregated model reads 2.

Our route: fix a row set **polynomial in (n, m)** , generate columns, carry *no* subtour constraints at all. Two such formulations follow.

BNP formulation 1 — PCF (position–configuration)

$y_C^k \in \{0, 1\}$: configuration C sits at position k . $w_t^k \geq 0$: insertion of tool t at k .

$$\min \sum_{t,k} w_t^k$$

$$(P) \sum_C y_C^k = 1 \quad \forall k \quad \text{one config. per position}$$

$$(Cov) \sum_k \sum_{C \supseteq T_j} y_C^k \geq 1 \quad \forall j \in J \quad \text{every job served}$$

$$(W) w_t^k \geq \sum_{C \ni t} (y_C^k - y_C^{k-1}) \quad t \in T, k \geq 2 \quad \text{charge each insertion}$$

$$(T) \sum_{k \geq 2} w_t^k + \sum_{C \ni t} y_C^1 \geq 1 \quad t \in U \quad \text{counting rows}$$

Rows are $O(nm)$ and fixed; the y_C^k are generated as columns.

Exact, but the relaxation is only the coverage bound

LP without (T) = 0 (the Tang–Denardo fractional pathology). LP with (T) = $|U| - b$. So PCF is a faithful *diagnostic instrument*, not a strong bound.

BNP: PCF' — killing the padding symmetry

PCF repeats configurations to fill all n positions; the repeats may sit anywhere, so *one* schedule has **many equal-cost encodings** — distinct nodes to the tree search.

$$(P') \quad \sum_C y_C^k \leq 1 \quad \forall k; \quad (G) \quad \sum_C y_C^{k+1} \leq \sum_C y_C^k \quad (k = 1, \dots, n-1)$$

with (Cov), (W), (T) unchanged.

- (G) forces used positions into a **prefix** $1, \dots, p$, empties into a suffix — one canonical encoding per schedule.
- (G) is *not* decorative: without it an interior empty position makes (W) misread a re-entry as a full reload and (T) misattribute the free first load.

Status

PCF' is exact; its LP *equals* PCF's ($|U| - b$). It removes symmetric integer encodings **but does not strengthen the bound**. Symmetry and bound are separate concerns — PCF' attacks the first, only PTF attacks the second.

BNP formulation 2 — PTF (position–transition)

Index *transitions*, not positions: $z_{C,C'}^k \geq 0$ is the step $k \rightarrow k+1$ that leaves C and enters C' , at cost $d(C, C') = |C' \setminus C|$.

$$\min \sum_{k=1}^{n-1} \sum_{(C,C')} d(C, C') z_{C,C'}^k$$

$$(S) \quad \sum_{(C,C')} z_{C,C'}^k = 1 \quad \forall k \quad \text{one transition per step}$$

$$(F) \quad \sum_{(C,C'): t \in C'} z_{C,C'}^k = \sum_{(D,D'): t \in D} z_{D,D'}^{k+1} \quad t \in T \quad \text{tool-by-tool flow consistency}$$

$$(Cov) \quad \sum_k \sum_{(C,C'): T_j \subseteq C} z_{C,C'}^k + \sum_{(C,C'): T_j \subseteq C'} z_{C,C'}^{n-1} \geq 1 \quad \text{every job served}$$

The first relaxation in this family to exceed $|U| - b$

The flow rows (F) chain the steps into one *coherent* walk — information a single position cannot carry. On a witness the PTF LP is 2.10 against a coverage bound of 2 (on the 6-ring, tight at $Z^* = 3$): its relaxation can exceed $|U| - b$.

BNP: column generation and robust branching

Both models: $O(nm)$ rows, but $\binom{m}{b}$ columns $\Rightarrow$ solve by *column generation* (restricted master + a pricer supplying most-negative reduced-cost columns). Driving it in a tree = *branch-and-price*.

Branch on a single column y_C^k ? No.

Under the padding symmetry it is ineffective, and it forces the pricer to carry a *growing forbidden-column list* — the oracle changes shape at every node.

Robust branching. Branch on the *presence* variables

$a_t^k = \sum_{C \ni t} y_C^k \in \{0, 1\}$ already in (W). Fixing $a_t^k = 0/1$ forbids/forces tool t at position k ; in the pricer this **shifts one dual weight** and leaves the oracle's *form* unchanged.

Why robust, and not Ryan–Foster here

Ryan–Foster (same/different-pair) branching exists for *set-partitioning* masters whose pricer *variable* branching would destroy. Our master is not of that type: the aggregated a_t^k branch keeps the pricer intact, so **RF would solve a problem we do not have**. Robust branching is the matched tool.

BNP: the two pricers (the algorithmic price of a bound)

PCF' pricer — one b -subset. With duals $\pi_j, \mu_{t,k}, \lambda_t, \dots$, the best column at position k maximises

$$\max_{|C|=b} \sum_{t \in C} \rho_t^k + \sum_{j: T_j \subseteq C} \pi_j, \quad \rho_t^k = -\mu_{t,k} + \mu_{t,k+1} + \lambda_t[k=1, t \in U]$$

choose b tools for per-tool rewards + a bonus per job whose *whole* requirement lands in C . This is a *Set-Union Knapsack* [Goldschmidt 1994; Crama 1994]. The branching dual enters **additively** in ρ_t^k — oracle unchanged. *Cheap*. **PTF pricer — a coupled pair $C \neq C'$.** The reduced cost is bilinear ($\sum_t [t \in C' \setminus C](1 - \lambda_t) +$ chaining terms), so the pricer must choose *two* b -subsets at once — linearised by McCormick, a MIP with $2m$ binaries. **Heavy**.

PCF': weak bound, cheap pricer. PTF: stronger bound, costly pricer. That trade-off is the entire BNP story.

BNP: results, and what the two failures mean

Validated range $n \leq 25$ (separate $n \geq 30$ run: both solve 0/96)

	solves	root bound = opt.	single-node solves
PCF'	94/319	93/94	15/94 (16%), median 89 nodes
PTF	72/291	—	71/72

- PCF' solves where its root bound already equals the optimum, but still needs search to find the integer solution: bound tightness and integrality are separate obstacles.
- PTF's stronger bound closes 71 of its 72 solves at the root node; its heavier pricer solves fewer instances overall.
- **Two-regime failure split** (PCF', on instances with known optima): of 225 failures, 73 are on bound-tight instances (a matter of speed) and 152 on bound-loose instances, which need a stronger relaxation rather than faster code.

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The datasets and why the campaign is run this way

Primary — of 411 instances, time limit 3600 s, all methods

Catanzaro Tabela 1C, Crama, Laporte T7. $n = 8\text{--}40$, $m = 5\text{--}60$, $b = 3\text{--}30$. The hard, size-varied suite; includes the flow model SSPMF.

Secondary — of 1,010 instances, time limit 600 s, SSPMF excluded

Laporte T3–T5. $n = 8/9/15$. Many small instances $\Rightarrow$ statistics. SSPMF is *too slow to build* at $n = 15$ under 600 s, so it is dropped *by design*.

Why re-run all baselines ourselves? One machine, one CPLEX generation, one KTNS-based metric. **Cross-paper wall-clocks are meaningless** — they mix hardware and solver vintages. This is benchmarking hygiene, and it is itself a finding (next slide: 2004 model $\approx$ 2015 model under modern CPLEX).

Why only selected instances run — and why primary looks poor

Early-stop protocol. Instances are ordered *easiest-first* by $(n, m, b, \text{density})$, identically for every method. A configuration is **retired after 8 consecutive non-optimal runs** — everything past that point is at least as hard.

- “Attempted” is therefore an *outcome*, not a design choice: a stronger method automatically reaches deeper. Budget saved on a dead arm is spent where it still informs.
- Totals use **uniform denominators** (the whole suite); every unattempted instance counts as *unsolved*. Nobody is flattered.

Why BNP scores 1/411 on primary — three honest reasons

(1) The prototypes are *validated* only for $n \leq 25$; primary is dominated by $n = 30\text{--}40$ (Catanzaro C/D, Crama), a separate run where they solve 0/96. (2) The pricers are *deliberately unaccelerated* — no warm columns, no stabilisation, no heuristic pricing: the prototype is a *measuring instrument* for the relaxation, not a race entry. (3) They are *bound-limited*: on loose instances no speed helps.

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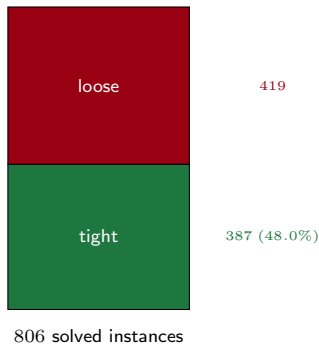
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The full table — both campaigns in context

solved (uniform denom.)	LSS'04	CATZ-F4'15	SSPMF'24	BBC	PCF'	PTF
primary, of 411	91	91	138	43	1	1
secondary, of 1,010	667	455	n/a	244	93	71

- On the primary suite SSPMF solves the most instances, consistent with its published result.
- Under the same modern CPLEX the 2004 model matches Catanzaro-F4 (2015) on the primary suite and solves more on the secondary; here the solver generation accounts for more of the difference than the formulation vintage.
- BBC solves fewer instances; on the 43 commonly solved instances it is about two orders of magnitude faster (previous slide), and it does not solve outside the bound-tight class.
- BNP contributes a diagnostic formulation (PCF', $LP = |U| - b$) and one whose relaxation can exceed $|U| - b$ (PTF).
- The largest instance solved by any method has $n = 15$; no method solves any instance at $n \geq 30$.

One number governs both campaigns



Among all 806 instances solved by *any* method, the coverage bound $|U| - b$ equals the optimum on 387 (48.0%) — an upper-biased estimate, since unsolved instances skew loose.

The split is roughly even

On the **tight** instances the exact methods close at the root and the practical heuristic tends to be exact; on the **loose** instances the gap opens and both campaigns stall. The 419 loose instances are the testbed for the planned polyhedral work.

BBC, BNP (PCF'/PTF) and the compact relaxations all meet at $|U| - b$: the configuration methods are bound-limited rather than implementation-limited.

Takeaways and next steps

- 1 **BBC:** Benders fits because loading is polynomial (KTNS $\Rightarrow$ exact cuts). Adding the coverage row $\theta \geq |U| - b$ recovers the tight half of the suite (6-ring: 0 cuts vs 443); the residual failures are bound-loose.
- 2 **BNP:** PCF' is the *diagnostic* (LP = $|U| - b$, symmetry removed); PTF's relaxation is the first in this family to exceed $|U| - b$, at the cost of a coupled-pair pricer; robust branching keeps pricing intact throughout.
- 3 **Joint:** one quantity, $|U| - b$, governs both, and the standard benchmark splits roughly evenly on each side of it.

Code, re-implemented baselines, and per-instance results

<https://github.com/shankar-n/JGP-SSP-Codes-Shankar>

- Tang & Denardo (1988) — the KTNS loading rule.
- Crama, Kolen, Oerlemans & Spieksma (1994) — NP-hardness; construction heuristics.
- Laporte, Salazar-González & Semet (2004) — LSS compact ILP and branch-and-cut.
- Catanzaro, Gouveia & Labbé (2015) — improved compact formulations (F4).
- da Silva et al. (2024) — multicommodity-flow model (SSPMF).
- Colares et al. (2026); Moreira (2026) — configuration/GTSP view and its correction.
- Burger et al. (2015) — colour-printing SSP; group-then-sequence in practice.
- Goldschmidt, Nehme & Yu (1994) — the Set-Union Knapsack (PCF' pricer).
- Barnhart et al. (1998); Lübbecke & Desrosiers (2005) — branch-and-price.